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Little Red Dots Should Turn Red Over Cosmic Time: A Predicted Subtype March and the Blue Dominance of the $z > 8.5$ Population

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ABSTRACT

Stacked spectroscopy of little red dots (LRDs) resolves an ordered continuum-subtype sequence, from extreme red (xLRD) to blue (bLRD) classes, that inverts to a monotonic sequence in the fraction of ionizing photons processed by dense nuclear gas. If this spectral diversity is a visibility sequence—one population seen at different stages of nuclear envelope growth—rather than distinct formation routes, the subtype mixture cannot be constant: it must evolve with redshift. We extract this prediction, with uncertainties, from a semi-analytic lifecycle model in which the envelope covering odds equal the squared engine-to-host continuum contrast. Averaged over realizations, the selected red fraction (xLRD+plusLRD) rises from 0.04 ± 0.05 at $z \simeq 9.5$ through 0.09 ± 0.04 and 0.23 ± 0.05 to 0.49 ± 0.09 at $z \simeq 4.5$ (plateau-consistent 0.48 ± 0.14 at $z \simeq 3.25$, an extrapolation below the calibration range), while blue-continuum LRDs are the largest class at $z > 8.5$ (0.61 ± 0.15 of the selected population), with xLRDs essentially absent there. The march is population physics, not selection: removing the survey selection strengthens it, and the rise to $z \simeq 4.5$ is monotonic in all realizations. It is testable now by splitting existing stacked and spectroscopic samples (predicted slope 0.10–0.14 per unit redshift between $z \simeq 7.5$ and 4.5). The same physics implies that the reported $z \simeq 9.5$ number density cannot currently constrain formation: the intrinsic UV-bright nuclear-active population exceeds it $\sim$34-fold but is blue and removed by the redness selection. A sample of ten classified $z > 8.5$ LRDs with a red majority would exclude the prediction at $\geq 99\%$ confidence even at its upper uncertainty edge.

Keywords: Active galactic nuclei (16) — Supermassive black holes (1663) — High-redshift galaxies (734)

1. INTRODUCTION

Little red dots—compact, red, frequently broad-line JWST sources at $z \gtrsim 4$ (Matthee et al. 2024; Greene et al. 2024; Kocevski et al. 2024, 2025)—show an ordered spectral-subtype sequence in stacked spectroscopy (Pérez-González et al. 2026). Inverting a dense-plus-diffuse two-zone photoionization model through the four stacks yields dense photon fractions of 0.981, 0.901, 0.738, and 0.086 from the xLRD through bLRD classes at nearly constant diffuse ionization parameter, with the population-level He II constraint (Sok et al. 2026) selecting a soft extreme-UV ionizing basis in that inversion. This ordered, one-parameter structure is what a *visibility sequence* looks like: a single population whose nuclear envelope intercepts a varying share of the ionizing output, rather than four unrelated formation routes. Dense-envelope models of individual LRDs (e.g., Naidu et al. 2025; de Graaff et al. 2025) have argued on independent grounds that envelope properties track the accretion state; spectral diversity has been mapped to varying black-hole-to-host luminosity ratios in a stationary framework (Barro et al. 2025); two LRDs have been observed transitioning into quasars at the phase exit (Fu et al. 2025); and low-spin halo models address the redshift evolution of LRD observability (Pacucci & Loeb 2025). The population-level consequence for the subtype mixture, however, appears not to have been stated as a quantitative, falsifiable prediction with a slope and uncertainties, which is the purpose of this Letter: if the dense photon share is set by the growth state of the nuclear envelope, and envelopes grow as nuclear reser-

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Table 1. Predicted selected subtype fractions (seed mean ± combined binomial and seed uncertainty). $N_{\rm eff}$ is the effective selection-weighted sample per bin.

z	xLRD	plusLRD	minusLRD	bLRD	$N_{\rm eff}$
9.5	0.00 ± 0.00	0.04 ± 0.05	0.35 ± 0.16	0.61 ± 0.15	21
7.5	0.00 ± 0.00	0.09 ± 0.04	0.52 ± 0.08	0.40 ± 0.08	52
5.5	0.00 ± 0.00	0.23 ± 0.05	0.57 ± 0.06	0.21 ± 0.05	265
4.5	0.01 ± 0.01	0.48 ± 0.09	0.46 ± 0.10	0.05 ± 0.03	56
3.25	0.01 ± 0.03	0.47 ± 0.13	0.47 ± 0.13	0.05 ± 0.06	19

NOTE—The $z = 3.25$ row is an extrapolation below the $4.5 < z < 6.5$ calibration window and shows the largest realization scatter.

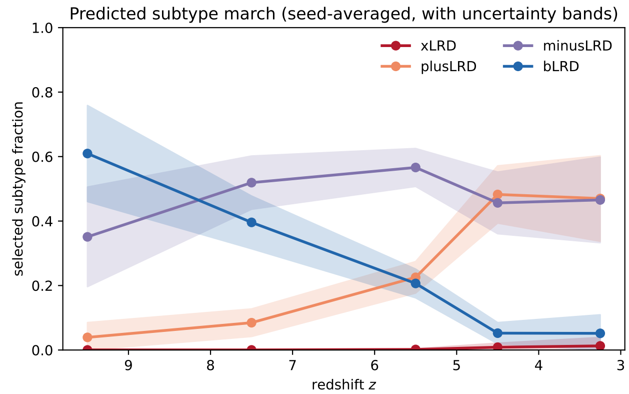

Figure 1. The predicted subtype march: seed-averaged selected subtype fractions versus redshift, with combined binomial and realization-to-realization uncertainties. Subtypes follow the published L_{5100}/L_{2500} boundaries; weighting is by selection probability times comoving weight. The $z \simeq 3.25$ bin lies below the demographic calibration range and is an extrapolation.

voirs feed while stellar hosts are still assembling, then the subtype *mixture must evolve with redshift*. A population seen earlier in its lifecycle must be bluer.

2. THE MODEL IN BRIEF

The prediction is extracted from a semi-analytic lifecycle model developed in a companion paper (Pandev 2026b, submitted). Halo histories, importance-reweighted to the Tinker et al. (2008) mass function, carry nuclear gas and black-hole reservoirs through an explicit five-state lifecycle with competing exit hazards; the observation layer assigns each source a porous nuclear reprocessor, a two-zone soft-EUV emission model built on thermal-balance Cloudy anchors (Chatzikos et al. 2023), and the published L_{5100}/L_{2500} subtype boundaries (1.8, 3.1, 6.3). The envelope covering C follows an equilibrium law: the covering odds equal the squared intrinsic engine-to-host continuum contrast,

$$\frac{C}{1-C} = x^2, \qquad x \equiv \frac{L_{\nu,\rm engine}(2500\,\text{Å})}{L_{\nu,\star}(2500\,\text{Å})}, \qquad (1)$$

applied with a frozen per-halo lognormal scatter of 0.35 dex. In the companion paper this law is selected against the stacked line ratios and the $z \leq 7.5$ number densities, its exponent is constructed through an explicit elimination program (momentum-limited envelope building against drag-limited clump removal; three alternative mechanisms rejected by the data), and the resulting photon partition matches the eight stacked [O III] ratios at least as well as an imposed closure while improving the reliable-bin demographic fit to $\chi^2 = 1.2$. Here only one property matters: covering—and therefore subtype—tracks the engine's contrast with the growing host.

3. THE SUBTYPE MARCH

Figure 1 and Table 1 give the march. The selected red fraction (xLRD+plusLRD) rises from 0.04 ± 0.05 at $z \simeq 9.5$ to 0.49 ± 0.09 at $z \simeq 4.5$—a many-sigma monotonic evolution—while the bLRD fraction falls from 0.61 ± 0.15 to 0.05 ± 0.03 and the median dense photon fraction climbs from 0.40 to $\simeq 0.85$. At $z > 8.5$, blue-continuum broad-line sources are the largest predicted class and xLRDs are essentially absent.

Two checks establish that this is population physics rather than a selection artifact. First, removing the selection weighting entirely *strengthens* the march (intrinsic red fraction $0.004 \rightarrow 0.7$ over the same interval, computed for the full catalogued population without a UV cut), and every subtype trend retains its sign: a transparent selection gate can rescale absolute fractions but cannot manufacture or erase a monotonic factor-of-$\sim$100 evolution that survives with no gate at all. Second, the rise from $z \simeq 9.5$ to $z \simeq 4.5$ is monotonic in all four random-seed realizations; the $z \simeq 3.25$ bin, with $N_{\rm eff} \simeq 19$ and the largest seed scatter, is reported as plateau-consistent and flagged as an extrapolation throughout.

The physical driver is simple: at high redshift most nuclear-active sources have young reservoirs and vigorous host star formation, so $x \lesssim 1$, envelopes are unbuilt, and continua are blue; as reservoirs feed and hosts age, x grows, covering saturates, and the population reddens. The march is the population-level shadow of individual sources evolving blueward-to-redward through the visibility sequence.

4. THREE TESTS

(i) The slope, with existing data. Between $z \simeq 7.5$ and 4.5 the red fraction should rise at 0.10–0.14 per unit redshift (range spans seed realizations and bin pairings). A redshift split of the existing 249-object stack sample (Pérez-González et al. 2026) and of spectroscopic compilations at $z \simeq 5.5$ tests this without new observations. One systematic must be controlled: fixed observed-frame filters sample different rest-frame wavelengths across this interval, and the surveys contributing at $z \simeq 4$ are not those contributing at $z \simeq 8$. The prediction concerns the *rest-frame* L_{5100}/L_{2500} classification; applying that classification uniformly (as the stacked-sample construction already does) mitigates the observed-frame drift, but the redshift dependence of the parent selection's color completeness does not cancel and should be forward-modeled or bounded before a measured slope is compared with the prediction.

(ii) Composition at $z > 8.5$. The model predicts bLRD plurality (0.61 ± 0.15) and xLRD absence at $z > 8.5$. The falsification threshold is quantitative: with a predicted red fraction of 0.04 ± 0.05, a sample of ten classified $z > 8.5$ LRDs containing a red majority (≥ 5 red) would exclude the prediction at $\geq 99.9\%$ confidence even adopting the $+1\sigma$ edge ($p_{\rm red} = 0.09$; binomial $P(\geq 5/10) = 1.0 \times 10^{-3}$), and at 2×10^{-5} at the central value.

(iii) Luminosity control. Because the driver is the engine–host contrast rather than luminosity, the dense-fraction–redshift anticorrelation should persist at fixed bolometric luminosity, distinguishing the march from a pure luminosity-selection trend.

5. THE $Z \simeq 9.5$ DENSITY IS A COMPLETENESS THERMOMETER

The same physics resolves the character of the reported high-redshift abundance. In the forward catalogue at $z \simeq 9.5$, the reported broad-selection density is 3.18×10^{-5} cMpc^{-3} (Rinaldi et al. 2026); the model's selected density is 5.0×10^{-6} (the factor 6.3 deficit), while its intrinsic $M_{\rm UV} < -18.5$ nuclear-active population totals 1.1×10^{-3}—a factor $\sim$34 above the reported value (a model-conditional reweighting; its headroom resides in sources assigned selection probabilities of 10^{-6}–10^{-3} by the model's analytic color gate). These sources are UV-bright ($M_{\rm UV} \simeq -19.4$), compact, and embedded, and are removed only by the redness selection. Rescaling that single gate within transparent limits moves the predicted $z = 9.5$ density from the factor $\sim$6 deficit to a factor $\sim$3 excess while moving the $z \leq 7.5$ bins by well under their errors: the bin measures color completeness—which Rinaldi et al. (2026) describe as essentially unconstrained at $z > 6$–7—rather than formation physics. The recent report of no evolution in the LRD number density across this range (Loiacono et al. 2026) is independently consistent with selection-dominated demography.

Table 2. Per-subtype selection at $z \simeq 9.5$ in the forward catalogue ($M_{\rm UV} < -18.5$; single fiducial realization, whereas Table 1 is seed-averaged).

subtype	intrinsic n [cMpc^{-3}]	selected n	pass fraction
xLRD	···	···	···
plusLRD	2.4×10^{-7}	1.4×10^{-7}	0.57
minusLRD	1.9×10^{-6}	1.5×10^{-6}	0.83
bLRD	1.1×10^{-3}	2.8×10^{-6}	0.0026

NOTE—No $M_{\rm UV} < -18.5$ xLRDs occur at this redshift in the fiducial realization. The selected shares implied here (3/34/63% for plus/minus/bLRD) agree with the seed-averaged Table 1 within its uncertainties.

Table 2 resolves an apparent paradox in these two claims: how can the *selected* $z \simeq 9.5$ sample be bLRD-dominated if the redness gate removes blue sources? Continuum subtype and the photometric V-shape gate are distinct axes: only 0.3% of intrinsic bLRDs pass the gate, but the intrinsic blue population is three orders of magnitude larger than the red one, so its thin passing tail still dominates the selected sample while the removed 34× headroom is also blue.

Two consequences follow: no additional formation channel is required by current number densities, and the informative measurement at $z > 8.5$ is compositional (test ii), not demographic. Injection/recovery completeness for *blue, compact* templates is the decisive survey product.

6. DISCUSSION

The march prediction is deliberately exposed. If LRD subtypes are distinct formation routes, the mixture can be redshift-independent and the prediction fails; the same null outcome would follow if the dense-fraction sequence reflects orientation, so a flat measured march would leave orientation and distinct-routes degenerate rather than selecting between them—the discriminating outcome is an *inverted* march (red fraction rising with redshift), which would falsify the visibility-sequence framework outright, or a confirmed march, which would establish LRD spectral diversity as an evolutionary sequence and calibrate the envelope-growth clock against cosmic time. The $z \lesssim 7$ portion of the prediction is insulated from the high-redshift completeness question; the $z > 8.5$ portion should be read jointly with Section 5.

This work is unfunded and was carried out independently. It made use of public data products from the JWST archive and of published catalogues by the teams cited herein. This research made use of NASA's Astrophysics Data System and the arXiv preprint repository.

Facilities: JWST (NIRSpec, NIRCam, MIRI)

Software: Cloudy (Chatzikos et al. 2023), numpy, scipy, matplotlib, astropy, colossus

DATA AND CODE AVAILABILITY

The extraction scripts and machine-readable march fractions with uncertainties accompany the companion paper as an MIT-licensed code archive (Zenodo DOI to be assigned at submission); the full forward-catalogue grid is a separate CC-BY-4.0 data record (Zenodo DOI to be assigned). Only public data products are used (Rinaldi et al. 2026; Ma et al. 2026; Pérez-González et al. 2026; Hviding et al. 2025; Sok et al. 2026).

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